

CASE STUDY

AI in Sustainable Polyurethane Systems Manufacturing and Supply Chain

connected procurement, supplier quality, inventory, formulation planning, production status, and sustainability records into one dashboard. This gave procurement, plant, quality, and leadership teams a shared view of supplier risk, stock health, batch progress, and sustainability exposure across the full operation.

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Problem

The client's polyurethane systems manufacturing operation was being challenged by raw-material variability, formulation sensitivity, shelf-life constraints, and sustainability reporting requirements across procurement, production, and quality. Unlike a simple distribution business, every batch depended on the right mix of polyols, isocyanates, catalysts, additives, and recycled or bio-based inputs, each with different lead times, minimum order quantities, storage rules, and compliance requirements. A small issue in supplier reliability, documentation, or material compatibility could quickly turn into a delayed batch, excess inventory, or a quality deviation.

At the same time, operational data was fragmented across spreadsheets, ERP records, supplier files, and plant logs. Procurement could not easily see which suppliers were at risk, planners could not reliably identify which lots should be used first, and production teams had limited visibility into batch health or traceability. Sustainability claims and circular-material documentation were also difficult to govern consistently, making it harder to prove renewable content, manage exceptions, and support audit-ready reporting.

How we solved it

Innovacio Technologies designed an AI-enabled control tower that connected procurement, supplier quality, inventory, formulation planning, production status, and sustainability records into one dashboard. This gave procurement, plant, quality, and leadership teams a shared view of supplier risk, stock health, batch progress, and sustainability exposure across the full operation.

We implemented supplier-risk scoring and procurement intelligence so the client could evaluate vendors by delivery reliability, COA completeness, batch quality, compliance status, and sustainability documentation. The platform also predicted late shipments, flagged missing approvals, and recommended alternate sourcing paths when risk thresholds were exceeded.

To improve planning, we added lot-level inventory optimization and shelf-life-aware replenishment so the team could prioritize usable materials, reduce expiry risk, and align raw-material decisions with customer order timing. The system also translated forecasted demand into reorder and consumption rules, helping planners balance working capital with batch continuity.



We further introduced batch-health scoring and anomaly detection to monitor process drift, highlight unusual process signatures, and surface likely root causes before a bad batch advanced too far. On the sustainability side, we built traceability and governance views that tracked renewable content, supplier declarations, waste streams, and circular-material usage in one controlled workflow.

The result was a more resilient polyurethane supply chain where supplier selection, inventory decisions, batch consistency, and sustainability governance were managed together, helping the client reduce waste, protect quality, and make faster, more confident decisions.

EXECUTIVE SUMMARY

A custom polyurethane systems house operates in a difficult middle ground: it is not a simple distributor, and it is not a standard commodity chemical plant. It must manage raw-material variability, formulation accuracy, supplier approvals, shelf-life constraints, customer-specific product specs, and sustainability demands at the same time. When those flows are managed with spreadsheets and disconnected ERP records, the result is late procurements, excess safety stock, production deviations, weak traceability, and avoidable customer delays.

Innovacio solved this by designing an AI-enabled control tower that connected procurement, inventory, formulation planning, supplier quality, production status, and customer commitments in one dashboard. The solution did not replace the plant's operating rules; it made those rules faster, more visible, and more consistent. That is exactly where the research points: AI in supply chain management is most valuable in demand forecasting, inventory control, supplier selection, and collaboration, while chemical-industry digitalization works best when it is tied to real process and governance constraints.

SUPPLIER QUALIFICATION AND RAW-MATERIAL PROCUREMENT

Business problem: The company needed to buy polyols, isocyanates, catalysts, fillers, colorants, and specialty additives from a large supplier base, but procurement was being



slowed by fragmented emails, manual approvals, and weak visibility into lead times, COA quality, and sustainability documentation.

Data problem: Purchase orders, supplier performance notes, COAs, non-conformance records, price lists, lead-time history, and sustainability declarations lived in separate systems. That made it difficult to see which suppliers were reliable, which materials were at risk, and which orders should be accelerated or substituted.

AI solution: Innovacio built a supplier-risk and procurement dashboard that scored vendors on on-time performance, batch quality, price volatility, compliance completeness, and carbon or renewable-content claims. The AI layer predicted likely late shipments, flagged missing documentation before approval, and recommended alternate supplier or alternate-material paths when risk thresholds were exceeded.

Why it matters: Research on AI in SCM repeatedly identifies supplier selection and collaboration as core AI use cases, while polyurethane circular-governance research shows the sector depends on strong vertical coordination and documentation discipline. Innovacio made that coordination visible in one control tower rather than across many spreadsheets.

Element	What the plant needed	What AI added	Typical KPI effect
Supplier scorecard	A clear view of supplier reliability and quality	Risk scoring from delivery, COA, defect, and dispute history	Fewer surprise failures
Approval workflow	Fast but controlled purchasing decisions	Auto-flagging of missing certificates and exceptions	Shorter PO cycle time
Material sourcing	The right input at the right time	Lead-time prediction and alternate-supplier suggestions	Lower expedite cost
Sustainability traceability	Proof of bio-based or recycled content	Document validation and claim tracking	Better audit readiness

Suggested implementation details:

- Start with the top 20 raw materials that create the most production risk, cost pressure, or sustainability exposure.
- Use supplier scorecards and document validation before adding more advanced optimization logic.
- Connect the dashboard to procurement approvals so the AI is embedded in the working process, not in a separate analytics tool.



- Re-score suppliers monthly and re-train risk rules when a major feedstock or shipping route changes.

INVENTORY, SHELF-LIFE, AND FORMULATION PLANNING

Business problem: The company needed to balance raw-material inventory, work-in-process, and finished-goods stock while handling shelf life, lot traceability, and product-specific formulations. Overbuying tied up working capital; underbuying stopped batches and delayed shipments.

Data problem: Inventory counts were not enough. The plant also needed demand history, formulation BOMs, batch dates, storage conditions, expiry windows, and customer order commitments. Without those signals in one place, planners could not see what should be consumed first, what was becoming obsolete, or what stock was at risk of expiring.

AI solution: Innovacio created a planning dashboard that forecasted demand, calculated optimal reorder points, monitored expiry risk, and prioritized lots by formulation compatibility and shelf-life urgency. The model did not just forecast volume; it translated demand into actionable replenishment and consumption rules.

Why it matters: AI in supply chain literature places inventory control and demand forecasting at the center of business value. For a polyurethane systems house, this means inventory is not just about quantity - it is about usable formulation options, batch timing, and the avoidance of expiring stock.

KPI	Before	After	Why it matters
Inventory accuracy	Manual and delayed	Near real-time visibility	Less hidden stock
Expiry risk	Detected after the fact	Flagged before consumption	Less waste
Reorder timing	Planner judgment only	AI-guided reorder point	Lower stockout risk
Batch allocation	First-in, first-out not always followed	Lot-aware consumption rules	Better traceability

Suggested implementation details:

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- Create a lot-level inventory model rather than relying only on item-level stock counts.
- Add shelf-life logic to every replenishment view so the system understands what can actually be used in the next production window.
- Use exception alerts for slow-moving, near-expiry, and nonconforming lots.
- Tie inventory decisions to customer order dates so the planner sees service impact, not just stock balances.

BATCH CONSISTENCY, QUALITY, AND PROCESS CONTROL

Business problem: In polyurethane manufacturing, small process drifts can alter viscosity, cure behavior, foam density, adhesion, or thermal performance. Those deviations can create scrap, rework, or customer complaints long after the batch leaves the plant.

Data problem: The useful signals already existed in the process: temperature, pressure, mixing ratios, hold times, reactor history, operator interventions, quality results, and non-conformance logs. The challenge was that these signals were not connected early enough to stop a bad batch from progressing.

AI solution: Innovacio built a production dashboard with anomaly detection and batch-health scoring. The system highlighted unusual process signatures, correlated them with past quality failures, and surfaced likely root-cause patterns so supervisors could intervene before the batch became a customer problem.

Why it matters: Chemical-industry digitalization research emphasizes process modeling, optimization, control, and safety as core uses for AI. The manufacturing AI literature also frames quality assurance and maintenance as especially strong application areas.

Control point	Data used	AI response	Operational gain
Mix ratio	Ingredient dosing and flow	Detects off-spec blend patterns	Less rework
Process temperature	Reactor and tank temperatures	Flags unstable thermal drift	More consistent quality



Cure window	Time to cure and set	Predicts delayed or early cure risk	Higher first-pass yield
Deviation log	Operator notes and alarms	Clusters repeated causes	Faster investigations

Suggested implementation details:

- Begin with the most sensitive formulations and highest-value customer products.
- Use batch-health scoring to give operators a simple decision rather than a complex model output.
- Link quality results back to supplier lots so procurement and production share the same root-cause view.
- Re-validate the model whenever a formula, raw material, or process setting changes.

SUSTAINABILITY, CIRCULARITY, AND SUPPLIER GOVERNANCE

Business problem: Sustainable polyurethane systems increasingly depend on bio-based feedstocks, recycled inputs, and transparent ESG reporting. That creates a new supply-chain burden: teams must prove what was sourced, where it came from, what fraction was renewable, and how it behaved in production.

Data problem: Sustainability claims, supplier declarations, recycled-content certificates, and waste records are often scattered across procurement, compliance, and operations teams. Without a governed data layer, the company cannot easily show progress toward circularity or compare material trade-offs.

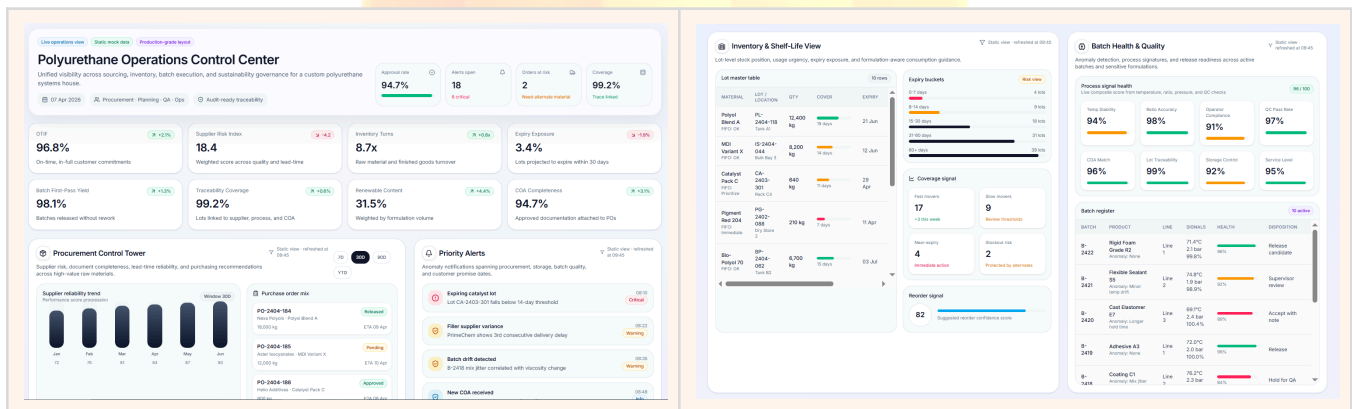
AI solution: Innovacio added a sustainability dashboard that connected sourcing data, product formulations, waste records, and supplier declarations. The platform tracked renewable content, supplier-document completeness, and circular-material usage while highlighting where sustainability goals conflicted with cost or lead-time risk.

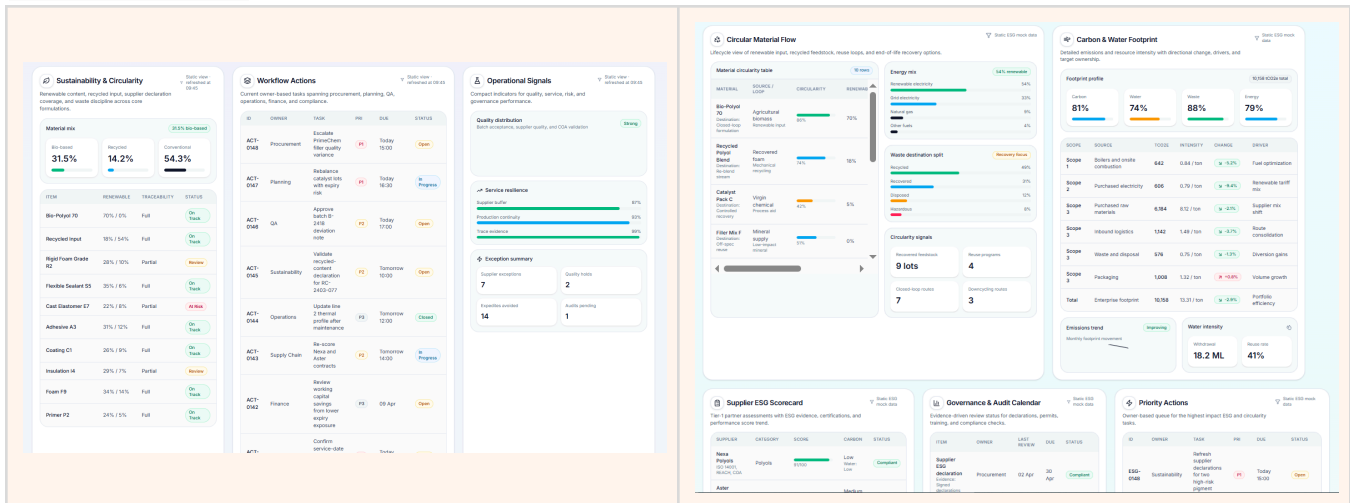


Why it matters: Research on bio-based polyurethane materials shows the transition to renewable feedstocks is promising but complex, while polyurethane circular-supply-chain work shows that governance across the value chain is essential. Innovacio turned that complexity into a measurable operating view.

Governance layer	AI contribution	Business outcome
Supplier claims	Checks completeness of declarations and certificates	Cleaner audit trail
Circular sourcing	Highlights renewable or recycled alternatives	Better sustainability decisions
Waste tracking	Connects waste streams to source lots and process conditions	Lower loss and better root-cause analysis
Reporting	Rolls procurement and production into one ESG view	Less manual reporting work

SCREENSHOTS:





FINAL TAKEAWAY

A custom polyurethane systems house succeeds when the supply chain, procurement team, inventory planners, and production team are operating from the same truth. Innovacio made that possible by turning supplier data, lot data, batch data, and sustainability data into a single AI-assisted operating system.

The result is a business that can buy more intelligently, carry less waste, protect batch quality, and prove sustainability claims with more confidence. That is the practical value of AI in this kind of chemical-manufacturing supply chain: not automation for its own sake, but faster and better decisions at every control point.

RESEARCH REFERENCES USED FOR THE CASE STUDY

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